

MAM2000W Module 2LA: Linear Algebra

Chapter 4

Some answers to the exercises

Section 4.1

1. The eigenvalues are -1 and 3 .

Basis for $W_{-1} = \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ and basis for $W_3 = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$.

2. $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

3. (a) Eigenvalue is 4 , basis for $W_4 = \left\{ \begin{pmatrix} 3 \\ 2 \end{pmatrix} \right\}$.

(b) This matrix has no real eigenvalues. It does have the two complex eigenvalues $\lambda_1 = \sqrt{3}i$ and $\lambda_2 = -\sqrt{3}i$. As a challenge, find the corresponding eigenvectors (they will have complex entries).

4. The eigenvalues are $3, 7$ and 1 .

5. Suppose that λ is an eigenvalue of A , with corresponding eigenvector $\mathbf{x}$. Let μ be a scalar.

$$\begin{aligned} (A - \mu I)\mathbf{x} &= A\mathbf{x} - \mu I\mathbf{x} \\ &= \lambda\mathbf{x} - \mu\mathbf{x} \\ &= (\lambda - \mu)\mathbf{x} \end{aligned}$$

So $\lambda - \mu$ is an eigenvalue of $A - \mu I$, with corresponding eigenvector $\mathbf{x}$.

6. The eigenvalues of A are $1, 2, 3$. Bases for the eigenspaces W_1, W_2, W_3 are (respectively)

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}, \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right\}, \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

(a) The eigenvalues of $A - 3I$ are $-2, -1, 0$, with the same eigenspaces as above.

(b) The eigenvalues of $A + 2I$ are $3, 4, 5$, with bases for the eigenspaces as above.

7. The eigenvalues are $-1, 1, -2$. Bases for the eigenspaces $W_{-1}, W_1,$

$$W_{-2} \text{ are (respectively) } \left\{ \begin{pmatrix} 2 \\ -1 \\ -1 \\ 0 \end{pmatrix} \right\}, \left\{ \begin{pmatrix} 2 \\ 3 \\ 1 \\ 0 \end{pmatrix} \right\}, \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}, \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

8. (a) $y = x, y = 2x$ (b) None (c) $y = 0$

9. $\dim(W_0) = 1$ or 2
 $\dim(W_1) = 1$
 $\dim(W_2) = 1$ or 2 or 3 .

10. Follow the instructions!

Section 4.2

1. (a) $3, 5$ (b) $\text{rank}(3I - A) = 1$ and $\text{rank}(5I - A) = 2$
(c) A is diagonalizable, because it has three linearly independent eigenvectors.
2. (a) $P = \begin{pmatrix} 4 & 3 \\ 5 & 4 \end{pmatrix}$ and $P^{-1}AP = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$
(b) $P = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $P^{-1}AP = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$
(c) A is not diagonalizable.
3. $A^{10} = \begin{pmatrix} 1 & 0 \\ -1023 & 1024 \end{pmatrix}$
4. Let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Recall that the j -th column of ${}_B[T]_B$ is $[T(\mathbf{v}_j)]_B$; if ${}_B[T]_B$ is a diagonal matrix it is also equal to an n -vector with zeros everywhere except possibly in the j -th position. Deduce from this that $\mathbf{v}_j$ is an eigenvector of A .

Section 4.3

1. (a) I_3 (b) I_3 (c) $\begin{pmatrix} 1 & -2 & 8 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & -2 & 8 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
2. (a) $P = \begin{pmatrix} 3 & -1 \\ 4 & 1 \end{pmatrix}$ and $D = \begin{pmatrix} 1 & 0 \\ 0 & -\frac{1}{6} \end{pmatrix}$
(b) $\lim_{n \rightarrow \infty} M^n = \begin{pmatrix} 0.4286 & 0.4286 \\ 0.5714 & 0.5714 \end{pmatrix}$ and $\lim_{n \rightarrow \infty} M^n \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} 0.4286(x_0 + y_0) \\ 0.5714(x_0 + y_0) \end{pmatrix}$.
3. 25% Red; 50% Pink; 25% White
4. (a) 35%; 37.5% (b) 40%
5. (a) It suffices to show that A and A^T have the same characteristic polynomial, so we check this:

$$\begin{aligned}
 \det(\lambda I - A) &= \det(\lambda I - A)^T \\
 &= \det((\lambda I)^T - (A)^T) \\
 &= \det(\lambda I - (A^T)).
 \end{aligned}$$

Make sure that you can justify each step.

(b) If A is a Markov matrix, the sum of the entries in each column is

1. So the sum of the entries in each row of A^T is 1. Then

$$A^T \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

(Try this out if you don't believe it!) Now this says that $\begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$ is an

eigenvector of A^T corresponding to the eigenvalue 1.

6. $\begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix} = b_1 \begin{pmatrix} 6 \\ 2 \\ 7 \end{pmatrix} e^{4t} + b_2 \begin{pmatrix} 0 \\ -7 \\ 1 \end{pmatrix} e^{-5t} + b_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} e^{2t}$ for $b_1, b_2, b_3 \in \mathbb{R}$.

7. $\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = 220 \begin{pmatrix} 2 \\ 1 \end{pmatrix} - 20 \begin{pmatrix} -3 \\ 1 \end{pmatrix} e^{-5t}$